Problem 1

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Problem 1 *Let f be a function given by: $f(x) = \left(\tan^{-1}(x)\right)^2$.*

(a) *State the domain of f. Write your answer in interval notation.*

Explanation. Domain of $f = (-\infty, +\infty)$

(b) *Find the critical points of f. Show your work!*

Explanation. Since

$$f'(x) = 2\tan^{-1}(x) \cdot \frac{1}{x^2+1} = \frac{2\tan^{-1}(x)}{x^2+1},$$

the only critical points of f are the points where $f'(x) = 0$.

$f'(x) = 0 \quad \equiv \quad \tan^{-1}(x) = 0 \quad \equiv \quad x = 0.$

Therefore, the function f has the only critical point at $x = 0$.

(c) *Show that* $\quad f''(x) = \dfrac{2 - 4x\tan^{-1}(x)}{(x^2+1)^2}.$

Explanation. $f''(x) = \left(\dfrac{2\tan^{-1}(x)}{x^2+1}\right)' = \dfrac{2\frac{x^2+1}{x^2+1} - 2\tan^{-1}(x)(2x)}{(x^2+1)^2} = \dfrac{2 - 4x\tan^{-1}(x)}{(x^2+1)^2}$

(d) *Use the **second derivative test** to determine whether critical point(s) of f correspond to local minimums, local maximums, or whether the test is inconclusive.*

Explanation. Since

$$f''(0) = \frac{2 - 4(0)\tan^{-1}(0)}{(0^2+1)^2} = 2 > 0,$$

it follows by the Second derivative test that the function f has a **local minimum** at $x = 0$.